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New Palaeomagnetic results from the northern Bushveld Complex of South Africa

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Abstract

Palaeomagnetic data have been acquired from seventeen sampling sites in the northern Bushveld Complex, Transvaal, South Africa. Twelve sites are situated in the main zone in the Potgietersrus area, while the remaining five are in the upper zone in the Villa Nora area. Specimens were subjected to alternating field and thermal demagnetization. Two mean directions were found. One direction represents the main zone and yields a pole at 1.6°N and 22.3°E, with associated polar errors d_p and d_m 14.7° and 15.9°. The second direction represents the upper zone in the northern Bushveld Complex with a virtual geomagnetic pole at 7.9°N and 359.3°E with d_p, d_m 44.4° and 54.6°. The apparent polar wander path indicates a sequential emplacement process for the Bushveld Complex, starting with the intrusion of the critical zone and ending with the emplacement of the upper zone.

1. Introduction

Several palaeomagnetic studies on rocks of the mafic sequence of the Bushveld Complex (Gough and van Niekerk, 1959; Hattingh, 1986a, 1986b, 1986c, 1989) have already established palaeomagnetic poles for the central Bushveld Complex. The northern Bushveld Complex comprises three distinct bodies: the Potgietersrus limb, the Villa Nora limb (Fig. 1) and a third limb in the Naboomspruit area obscured by Waterberg sediments. This investigation was initiated to provide the first palaeomagnetic data for these limbs and so help to delineate the genetic relation between the northern and central parts of the Bushveld Complex, as well as the relation between the sedimentary rocks and the layered mafic sequence.

2. Geology of the northern Bushveld Complex

The Potgietersrus limb of the northern Bushveld Complex is just over 100 km in length (Gain and Mostert, 1982) (Fig. 1). It is roughly triangular in outline and consists of a central felsic portion with an outer rim of basic to ultramafic rocks. Younger sedimentary cover obscures one side of the triangle and much of the centre. This limb is truncated in the south by the northeast-trending Zebedela fault that brings Karoo sediments into juxtaposition with the layered mafic rocks. Quaternary and Waterberg sedimentary rocks overlay the Potgietersrus limb to the north. To the west of the limb, the roof rocks are exposed, and consist of a variety of acid rocks and metasediments. The Archaean granites which compose the floor are exposed to the east.

The informal scheme of subdivision, based on a zonal nomenclature defined by Wager and

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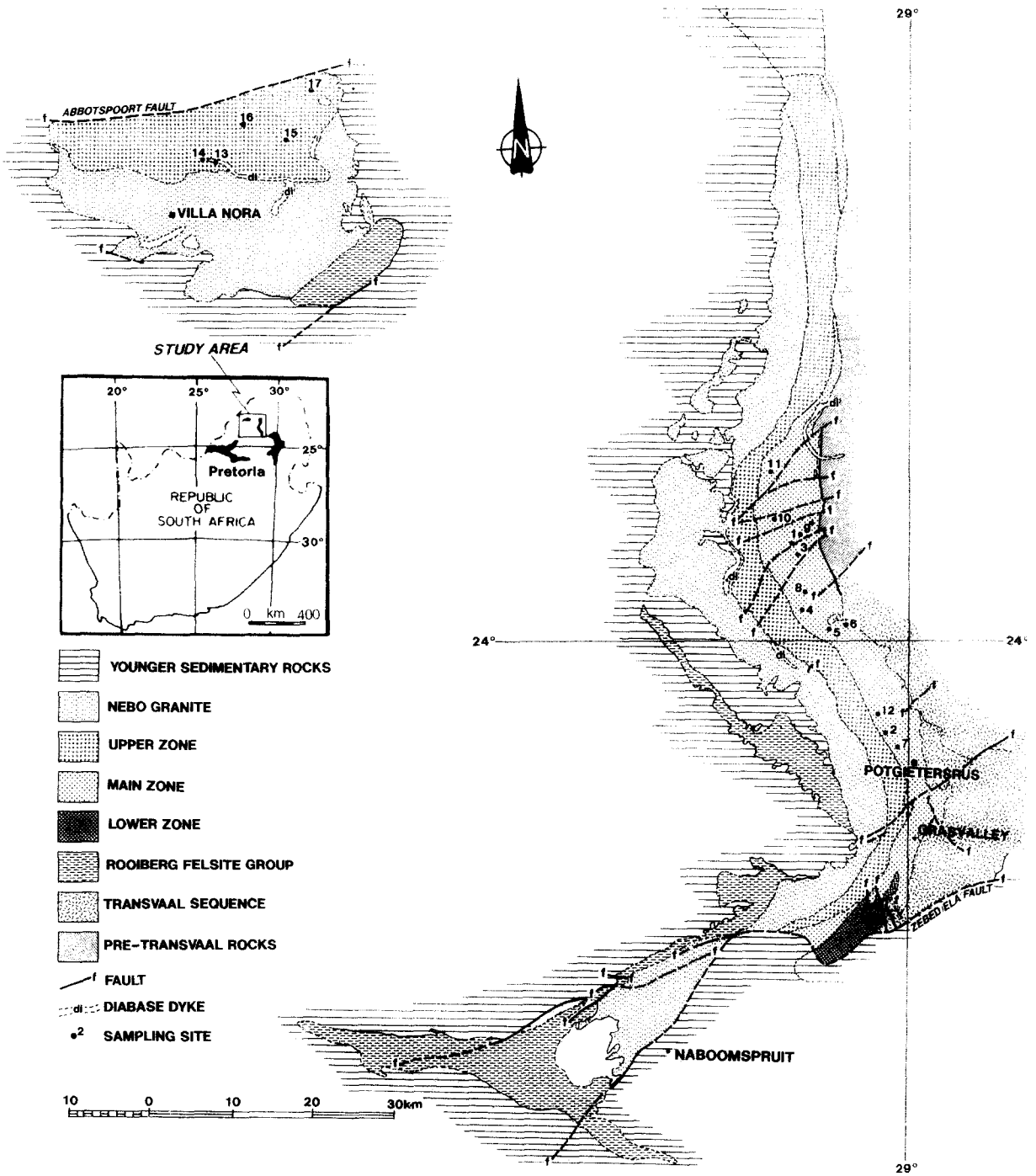


Fig 1. Geological map of the northern Bushveld Complex, showing sampling sites.

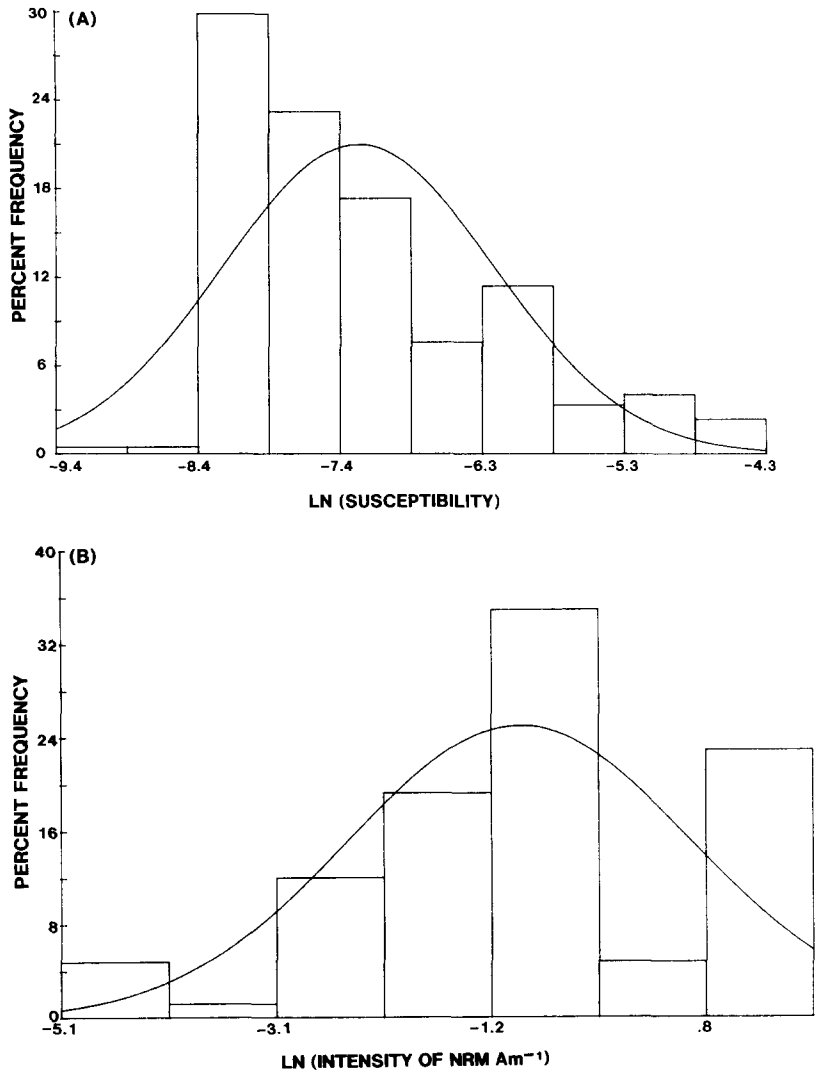


Fig 2. Histogram plots of susceptibility values (A), and intensity of NRM (B), from the main zone. The curved line is a normal distribution overlay.

Brown (1968), is retained. The lower zone is best represented in the Grassvalley–Volspruit area to the south of Potgietersrus, where it consists of a 1600-m thick sequence of pyroxenites and peridotites (Hulbert and Von Gruenewaldt, 1982). A limited thickness of 400 m of the critical zone occurs in the Potgietersrus area and this has its best exposure south of Potgietersrus, and consists of pyroxenites, melanorites and norites (Van der Merwe, 1976). The sequence of rocks constituting the main zone at Potgietersrus differs from the rest of the Bushveld Complex and is made up of gabbro, norite and troctolite with anorthosite and pyroxenite layers (Hulbert and Von Gruenewaldt, 1982). The best exposure is in the central portion of the Potgietersrus limb where it at-

Table 1
Site statistics of magnetization vectors from the main zone

Site	NRM directions					Directions after AF demagnetization					Directions after AF demagnetization with layering horizontal					
	N	D	I	α_{95}	κ	N	D	I	α_{95}	κ	dip	dip direction	D	I	α_{95}	κ
1	7	78.3°	63.8°	10.6°	33.2	6	77.2°	56.8°	10.7°	40.4	29°	286.5°	19.1°	74.1°		
2	6	355.8°	59.4°	53.2°	2.5	4	352.2°	70.2°	55.2°	3.7	25°	224.5°	276.3°	70.1°		
3	8	36.2°	62.4°	45.4°	2.4	6	59.6°	77.6°	19.5°	12.8	22°	237.5°	234.8°	80.4°		
4	7	3.6°	83.8°	15.6°	16.0	5	57.8°	64.2°	20.2°	15.3	15°	243.0°	50.9°	79.0°		
5	7	346.0°	59.2°	39.6°	3.3	7	18.6°	57.8°	16.6°	22.2	15°	243.0°	354.8°	66.3°		
6	7	348.1°	49.2°	44.6°	2.8	5	22.2°	45.4°	12.5°	38.6	15°	243.0°	8.6°	55.6°		
7	7	12.7°	48.0°	30.0°	5.0	5	17.7°	48.8°	8.4°	83.2	20°	258.0°	353.0°	55.0°		
8	5	29.7°	41.0°	21.8°	13.3	6	62.9°	77.7°	24.9°	8.3	25°	239.0°	235.2°	77.3°		
9	7	39.7°	70.5°	8.9°	47.4	7	42.9°	72.3°	3.5°	305.6	21°	262.0°	319.7°	76.9°		
10	6	45.3°	68.2°	26.0°	7.6	6	33.0°	80.3°	17.0°	16.3	16°	268.0°	305.9°	77.0°		
11	6	81.8°	68.2°	29.8°	6.0	5	41.0°	63.6°	4.6°	280.6	17°	270.0°	3.0°	70.3°		
12	7	30.5°	53.7°	5.8°	107.6	5	29.9°	50.4°	7.0°	120.6	25°	224.0°	10.0°	73.5°		
Mean direction		24.3°	63.8°	10.0°	19.5		35.2°	65.6°	8.5°	27.0			345.1°	76.3°	8.6°	26.3

N is the number of core samples; D is the declination of the magnetization vector, measured from north in a clockwise direction; I is the inclination of the magnetization vector, downwards positive, upwards negative; α_{95} is the semi-angle cone of 95% confidence; κ is the estimate of the precision parameter K; dip and dip direction are those estimated for the igneous layering at the sampling site.

Table 2
Site statistics of magnetization vectors from the upper zone

Site	NRM directions					Directions after thermal demagnetization					Directions after thermal demagnetization with layering horizontal					
	N	D	I	α_{95}	κ	N	D	I	α_{95}	κ	dip	dip direction	D	I	α_{95}	κ
13	5	343.7°	4.2°	56.7°	2.8	5	347.6°	48.3°	9.3°	68.9	40°	208.5°	288.2°	63.5°		
14	4	301.9°	53.0°	46.4°	4.9	6	280.8°	72.6°	33.5°	5.1	40°	208.5°	231.1°	42.3°		
15	4	218.9°	−20.3°	31.7°	9.4	–	–	–	–	–	–	–	–	–		
16	6	260.5°	60.8°	30.6°	5.7	7	287.0°	64.9°	34.8°	4.0	65°	174.5°	202.0°	32.0°		
Mean direction		284.0°	36.0°	79.9°	1.0		314.6°	65.6°	33.6°	14.5			229.8°	50.6°	50.5	7.0
				($R < R_c$)												

Symbols as in Table 1.

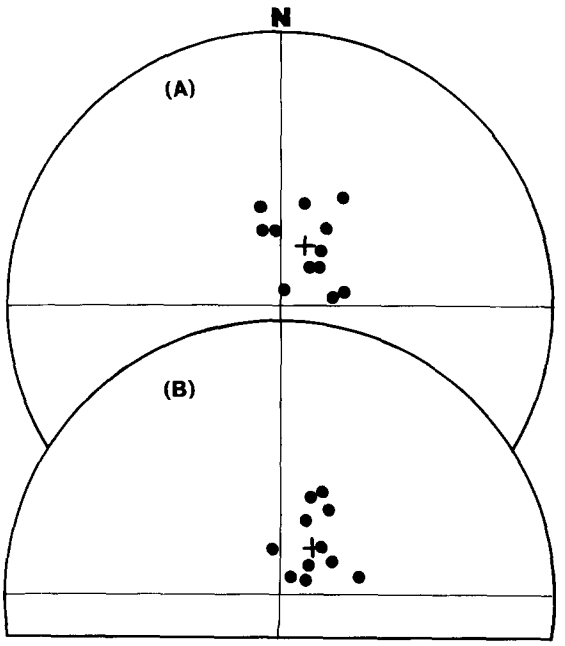


Fig 3. Stereographic projection of NRM site mean directions for the main zone (A) and (B) site mean directions after AF demagnetization. ● represents a positive inclination and ○ negative inclination. Group mean direction is indicated by +.

tains a maximum thickness of 2380 m. The upper zone occurs virtually along the entire strike of the limb, develops a thickness of 1100 m and contains magnetite layers, magnetite gabbro, gabbro and anorthosite cyclic units.

Further west, rocks of the upper zone constitute the Villa Nora limb (Fig. 1) whose northern flank is bounded by the Abbotspoort fault (McCourt, 1983). This has downthrown the layered sequence against the granite basement. The upper zone has an easterly strike, and dips of up to 60° to the south and southeast have been recorded (Grobler and Whitfield, 1970). The main rock types in this limb are leuconorites, leucogabbro and anorthosite.

The third limb is obscured by Waterberg sediments and is situated southeast of the Potgietersrus limb. Gravimetric data (Smit et al., 1962; Du Plessis, 1983) indicate that this limb runs in a northwest direction from Naboomspruit in the south to the Mogol River in the northwest. Van der Merwe (1976) postulated an eastward dip for this body and it follows from this that the three limbs together form a funnel-shaped structure.

3. Sampling and measurements

Samples were taken at seventeen sites, twelve of which are located in the main zone of the Potgietersrus limb and the remaining five in the upper zone of the Villa Nora area (Fig. 1). The paucity of the sampling sites is explained by the fact that the northern Bushveld Complex is situated in virtually flat country, where large parts of the succession are covered by turf. Sampling was thus restricted to stream beds where possible, or low-lying rock outcrops, as no road cuttings or quarries were available.

Cores (25 mm × ~200 mm), oriented by sun compass, were obtained with a hand-held, petrol-driven drill. The number of cores drilled at each site varied between five, where the outcrop was limited, and eight. Each core was cut into 25 mm long specimens.

The laboratory measurements were done in the palaeomagnetic laboratory of the Geological Survey of South Africa in Silverton, Pretoria.

Initially, bulk susceptibility measurements were made on each specimen from the main zone using a low-field susceptibility bridge attached to the Digico system (Fuller, 1967). The natural remanent magnetization (NRM) of specimens was measured using a Digico "Complete Results" fluxgate spinner magnetometer which is designed to measure rocks with an intensity of magnetism as low as 1×10^{-3} A m^{-1}.

Next, stepwise alternating field (AF) demagnetization was done on representative specimens from the main and upper zones on an AF demagnetization unit similar to that described by McElhinny (1966). Between 15 and 23 steps were performed on each specimen, according to the complexity of the remanence. The process was terminated when the AF limit (135 mT) was reached, or when the remaining magnetization became unstable. Other specimens from each site were then subjected to stepwise thermal demag-

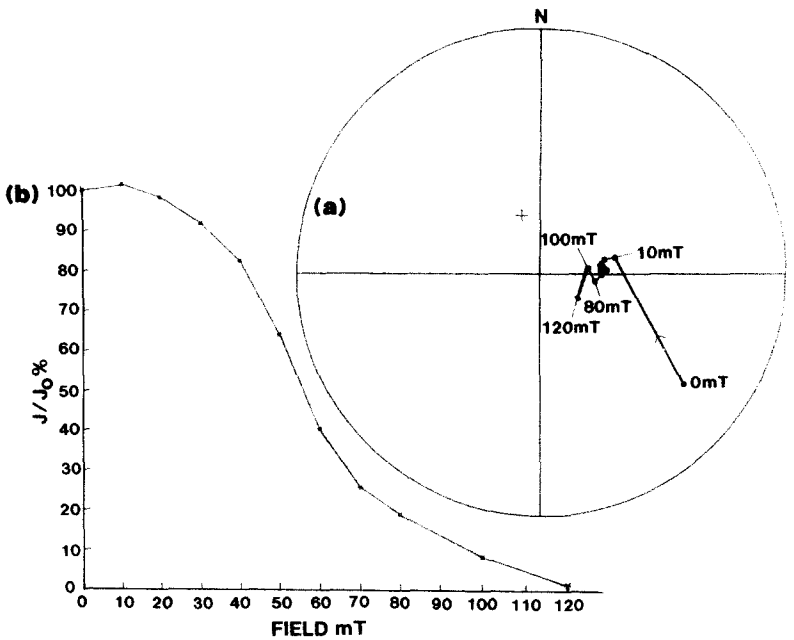


Fig 4. (a) Directional response to AF demagnetization and (b) normalized intensity response curve. Specimen B3/6E, main zone.

netization using, initially, steps of 100°C but reducing to steps of 10°C as the Curie temperature of magnetite was approached.

Directions of components of magnetization for the specimens were found using Zijderveld orthogonal projections (Zijderveld, 1967) and principal component analysis (PCA) (Kirschvink, 1980).

Lastly, polished thin sections were studied with a petrographical microscope. Furthermore, the opaque minerals were identified using reflected light. Where grain sizes were too small for positive identification, identification was attempted with a scanning electron microscope.

4. Results from the Main zone

The magnetic susceptibility values did not follow a normal, a lognormal or a three parameter lognormal (Rendu, 1981) distribution. However, a chi-square test applied to the data at the 0.05 significance level indicated that a lognormal distribution is the closest approximation. The estimated mean susceptibility of 1315×10^{-6} SI was calculated using Sichel's t estimator (Rendu, 1981), from 422 specimens from the main zone. A histogram plot of the susceptibility values for the main zone is shown in Fig. 2A. The intensities of NRM approximate a lognormal distribution (Fig. 2B). The estimated mean intensity, using Sichel's t estimator (Rendu, 1981), for the specimens from the main zone is 1644×10^{-4} A m^{-1}. From this the Koenigsberger ratio was calculated to be 52.5, which indicates that the remanent magnetization is dominant over the induced component of magnetization. This implies that any interpretation of the magnetic anomaly of the Potgietersrus limb must take the remanent magnetization into account.

Specimen NRM directions within all twelve sites are well grouped as are the twelve site mean directions. Table 1 gives the NRM site mean di-

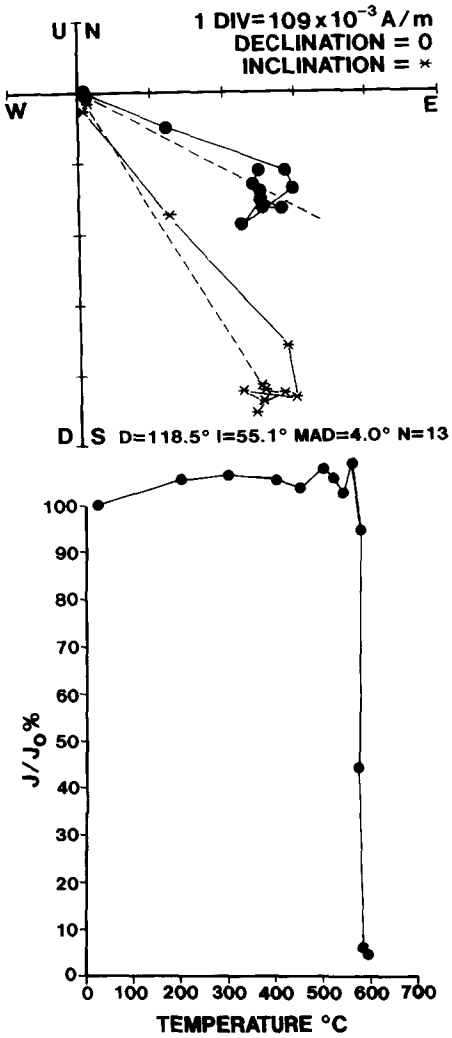


Fig 5. Response to thermal demagnetization. D=declination; I=inclination; MAD=maximum angular deviation, N=number of data points used for fit, dashed line=vector fitted with PCA.

rections (see also Fig. 3). The group statistics are: $n=12$; $D=24.3°$; $I=63.8°$; $\alpha_{95}=10.0°$; $\kappa=19.5$.

It was found that alternating field (AF) demagnetization best isolated the remanent magnetization for specimens from the main zone. One specimen from each core was AF demagnetized. These results were analyzed with Zijderveld diagrams and principal component analysis. The site mean magnetization directions

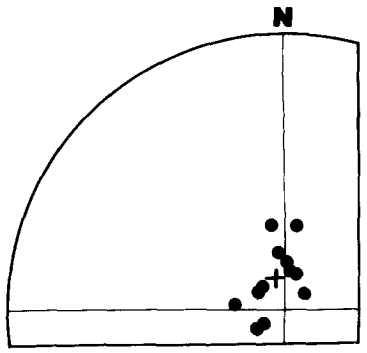


Fig 6. Stereographic projection of magnetization directions of sites in the main zone after AF demagnetization with the layering in the horizontal position.

obtained after AF demagnetization are listed in Table 1 and plotted in Fig. 3. AF demagnetization succeeded in improving the grouping of the magnetization directions of specimens from some of the sites. Statistics of the magnetization directions of sites after AF demagnetization are: $n=12$; $D=35.2°$; $I=65.5°$; $\alpha_{95}=8.5°$; $\kappa=27.0$.

A typical directional response to AF demagnetization is shown in Fig. 4. A low-coercivity component of magnetization is quickly removed, whereupon the magnetization direction remains almost constant. This plus intensity response curve revealed a stable magnetization which persists up to demagnetizing fields of approximately 115 mT (Fig. 4). Similar magnetizations, carried by high-coercivity grains have been observed in the Modipe Gabbro by Evans et al.(1968), in the Lambertville Diabase by Hargraves and Young (1969) and in the Bushveld Complex by Hattingh (1986a,b,c, 1989).

The intensity response curves of the specimens from the main zone after stepwise thermal demagnetization displayed a discrete blocking-temperature spectrum with two Curie points: a secondary component at around 475°C and a very well defined primary component at 578°C (Fig. 5). The Curie temperature of 475°C could be associated with haematite–ilmenite, while the well defined Curie temperature at 578°C confirms that magnetite is the primary source of remanent magnetization in these specimens.

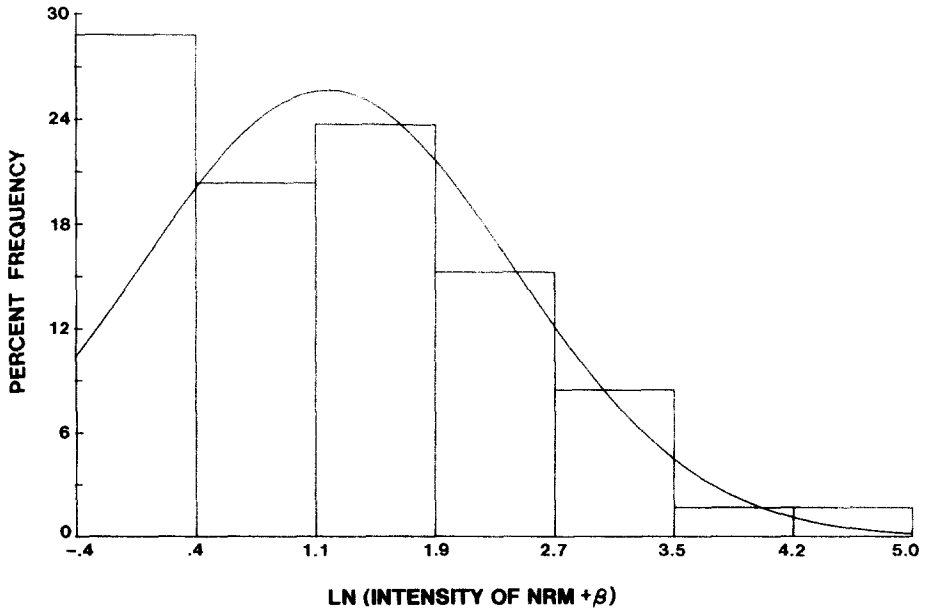


Fig 7. Histogram of intensity of magnetization of specimens from the upper zone. β is the additive constant for the three-parameter lognormal distribution.

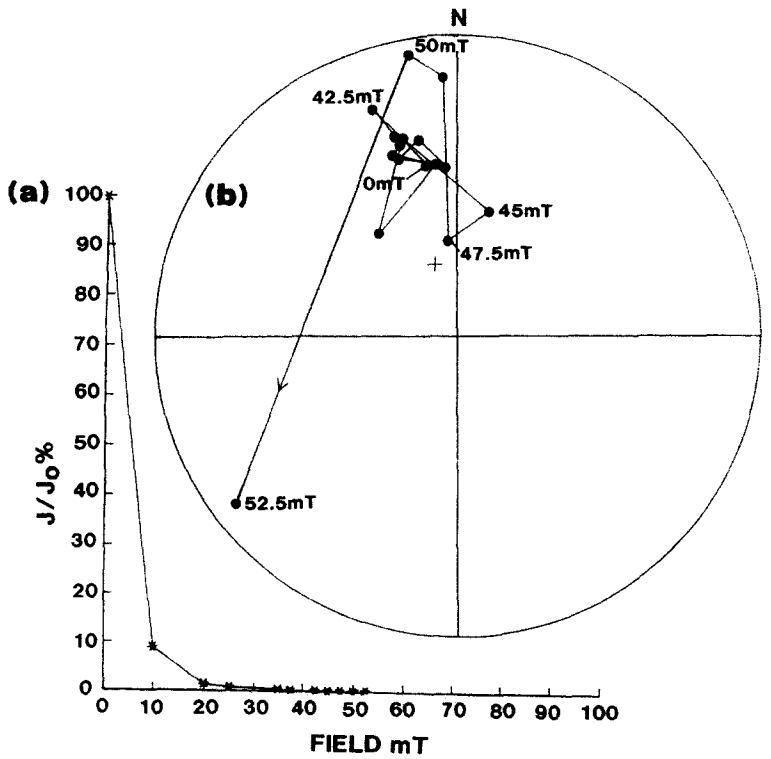


Fig 8. (a) Directional response of a specimen to AF demagnetization and (b) normalized intensity response curve, specimen B14/1B, upper zone.

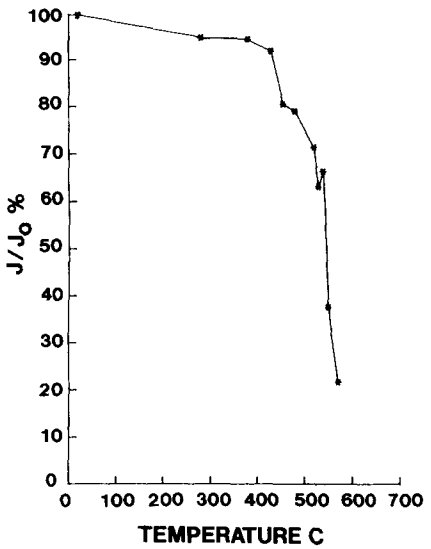


Fig 9. Thermal demagnetization intensity response curve for specimen B13/2A in the upper zone.

A fold test (Graham, 1949) was applied to the magnetization directions obtained by AF demagnetization (Table 1). The mean corrected direction is: $D=345.1°$; $I=76.3°$; $\alpha_{95}=8.6°$; $\kappa=26.3$. The dip directions used for the fold test (Table 1) fall within a narrow azimuth, therefore the corrections applied are similar, so that the chance of a significant improvement is small (Fig. 6). However, Hattingh (1986b,c) found that the main zone of the central Bushveld Complex had acquired its magnetization with the igneous layering in a horizontal position, this, with the fold test results obtained here, indicate that the main zone of the northern limb had also acquired its magnetization with the igneous layering in a horizontal position.

Specimens from the main zone are characterized by the presence of magnetite, ilmenite and various sulphide minerals. Both discrete magnetite grains, and magnetite needles within the pyroxene and plagioclase crystals, are present. The Curie temperature of 578°C, as well as the previously described "hard" magnetization suggest that the needles are in fact magnetite and are the magnetization carriers.

5. Results from the upper zone

The magnetization intensities approximate a three-parameter lognormal distribution. An additive constant β was calculated with the aid of a Geostokos geostatistical package (Clark, 1987). A histogram plot of the intensities of the specimens is shown in Fig. 7. The estimated mean intensity of the NRM of the specimens is 7982×10^{-3} A m^{-1}. This value is much higher than that obtained for the main zone. The NRM specimen directions within four sites were grouped,but no consistency exists between sites (Table 2).

Alternating field (AF) demagnetization was not effective at isolating the primary magnetization directions, as the specimens acquired rotational remanent magnetization (RRM) during demagnetization (Collinson, 1983). An attempt was made to minimize the RRM by using the method described by Hillhouse (1977). Fig. 8 shows the typical behaviour of the specimens from the upper zone during AF demagnetization. The intensity response curve as shown on Fig. 8 is typical for specimens in the upper zone with the shape of the curve indicating that the bulk of the magnetization is carried by low-coercivity grains which could be multi-domain (Dunlop, 1981; Hattingh, 1986b).

Stepwise thermal demagnetization proved more effective at removing secondary components for specimens in the upper zone. One specimen from each core was thermally demagnetized. Site 14 exhibited dual polarity, so that the site was divided into 14A and 14B. After demagnetization, sites 13, 14A, 14B and 16 grouped significantly at the 95% confidence limit (Watson, 1956). The magnetization direction for site 14A is reversed in relation to the rest of the sites. The mean direction for site 14A was inverted and statistically combined with site 14B to give a single site mean direction, site 14. The intensity response curves to thermal demagnetization indicate that two main components of remanence exist, a secondary low-blocking-temperature component associated with maghemite at approximately 400°C, and a primary high-block-

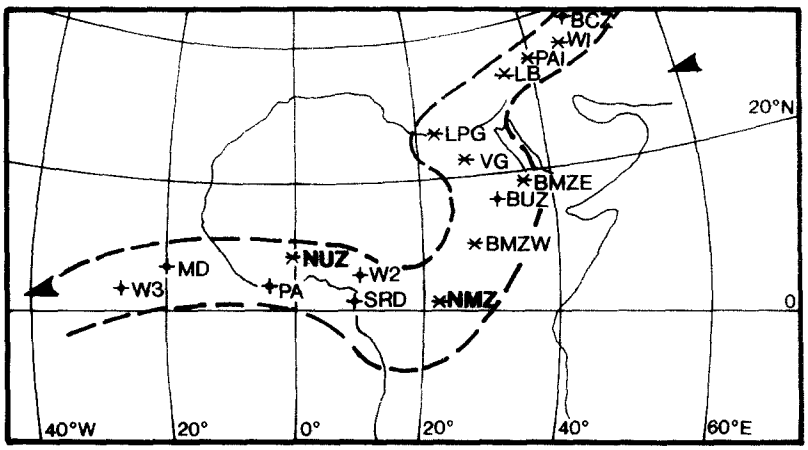


Fig 10. APW path with selected pole positions shown for southern Africa ~2070–1700 Ma. *BCZ* (2054±11 m.y.) = critical zone Bushveld Complex; *W1* = Waterberg sediments; *PA1* (2047±11 m.y.) = Phalaborwa pyroxenite; *LB* (1840±276 m.y.) = Losberg Gabbro; *LPG* (1980±39 m.y.) = Limpopo gneisses; *VG* (2006±47 m.y.) = Vredefort Granophyre; *BMZE* (2049±47 m.y.) = main zone, eastern Bushveld Complex; *BUZ* = upper zone Bushveld Complex; *BMZW* = main zone, western Bushveld Complex; *NMZ* = main zone, northern Bushveld Complex (this study); *NUZ* = upper zone, northern Bushveld Complex (this study); *SRD* (1876±68 m.y.) = Sand River dykes; *W2* = Waterberg sediments, *PA* (2012±19 m.y.) = Phalaborwa syenites and dolerites; *MD* = Mashonaland dolerites; *W3* = Waterberg sediments; asterisk = *N*-seeking magnetizations, + = S-seeking magnetizations.

ing-temperature component carried by magnetite at 578°C (Fig. 9).

A mean direction after demagnetization for the upper zone was thus calculated from three site mean directions. Table 2 lists the directions and relevant statistics for the upper zone after demagnetization. The mean group direction is: $D=314.6°$; $I=65.6°$; $\alpha_{95}=33.6°$; $\kappa=14.5$.

A fold test (Graham, 1949) was then applied to the magnetization directions obtained by thermal demagnetization (Table 2). With the layering in the horizontal position the statistics degenerated with the corrected mean direction: $D=229.8°$; $I=50.6°$; $\alpha_{95}=50.5°$; $\kappa=7.0$. Here again the dip directions are similar so that the same circumstances exist as those for the main zone. Due to the above limitations the fold test can be regarded as inconclusive regarding the position of the igneous layering during the acquisition of the magnetization.

6. Conclusions

This study succeeded in determining a mean magnetization direction for the main zone of the northern Bushveld Complex from which a pole (NMZ) position at 1.6°N and 22.3°E (igneous layering horizontal) was calculated (Fig. 10). The polar error (d_p, d_m) is 14.7° and 15.9°, respectively. Thermal demagnetization of specimens in the upper zone yielded stable and consistent magnetization directions from three sites. From this a virtual pole (NUZ) was calculated which is situated at 7.9°N and 359.3°E (Fig. 10), with the polar error (d_p, d_m) 44.4° and 54.6°, respectively.

Although poles determined in this study are compatible with the APW path determined by previous studies (McElhinny and McWilliams, 1977; Hattingh, 1986a,b,c, 1989), they suggest a more complex APW path than the broad swathe previously indicated by these authors. With poles

from this study and poles from previous studies it is possible to define a much more complex APW path (Fig. 10). This would suggest that the movement of the African continent during the 2100–1800 Ma time span was more intricate than previous palaeomagnetic data suggest. The pole for the main zone is different to those poles found by Hattingh (1986a,b,c) for the central Bushveld Complex, indicating that the emplacement of the central Bushveld Complex and the northern Bushveld Complex did not take place simultaneously.

From the APW path the critical zone of the central Bushveld Complex was the first phase in the emplacement of the Bushveld Complex. Van der Merwe (1976) postulated that the emplacement of the critical zone in the Potgietersrus limb was the first event in the genesis of the northern Bushveld Complex. Unfortunately no samples are available from this zone to substantiate this.

The second phase was the emplacement of the main zone of the central Bushveld Complex followed by the emplacement of the main zone of the northern Bushveld Complex (Fig. 10). The palaeomagnetic pole for the upper zone suggests that this is the last phase in the Bushveld Complex's genesis.

These different phases do not suggest a single magma pulse, but a series of related events that are genetically interlinked. This is in agreement with Sharpe (1981) who postulated that more than one magmatic series gave rise to the mafic layered sequence of the Bushveld Complex.

The Waterberg events were only completed after the emplacement of the northern Bushveld Complex (Fig. 10).

The pole found for the upper zone further indicates that no major tectonic movement occurred between the Potgietersrus and the Villa Nora limbs. The structural shape postulated for the northern limb has been based on gravimetric and magnetic interpretations (Van der Merwe, 1976; Du Plessis, 1983). The high Koenigsberger ratio calculated in this study indicates that previous interpretations may be inaccurate. Reinterpretation of magnetic data taking the remanent magnetization into account may well yield a different structure for the northern Bushveld Complex.

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